Master program
” Advanced Analytics for Business”

Advanced Visual Text Analytics – Natural Language Processing

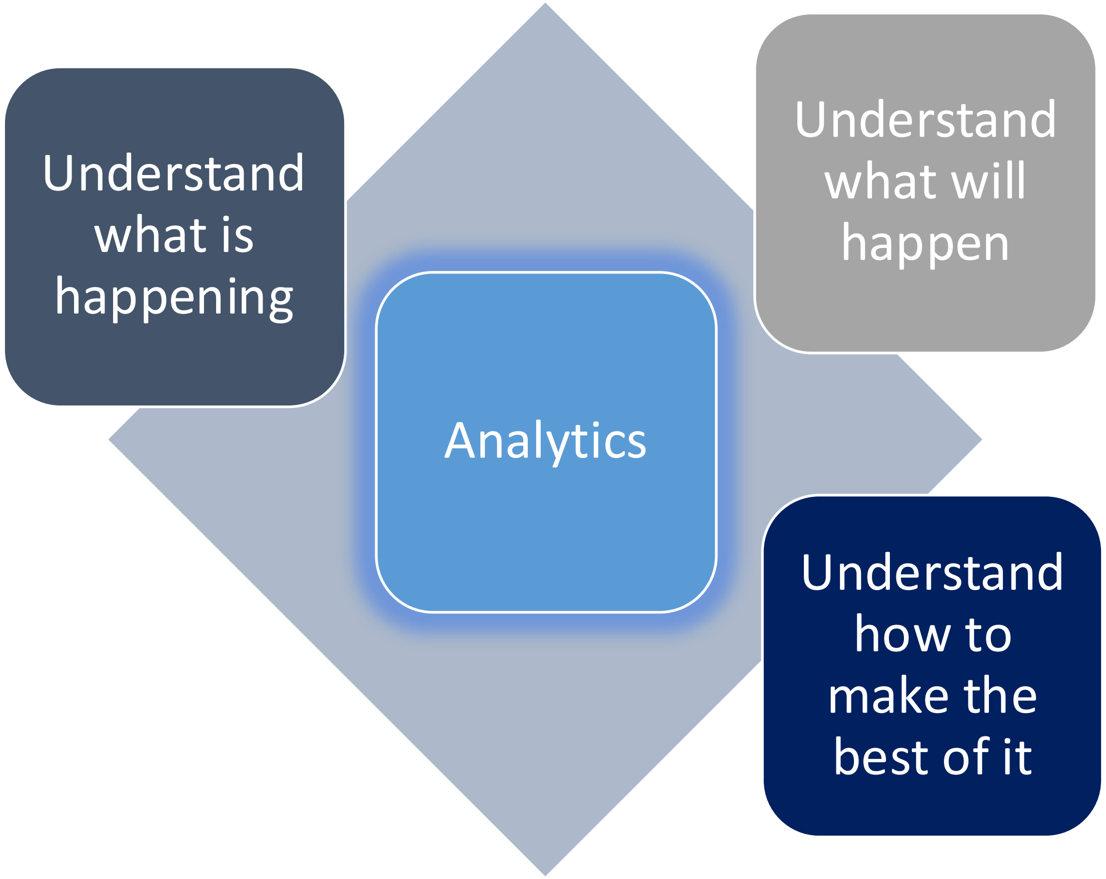
Universitatea Națională de Știință si Tehnologie Politehnica București
Facultatea de Automatică și Calculatoare

Course 1
Introduction

Lecture content

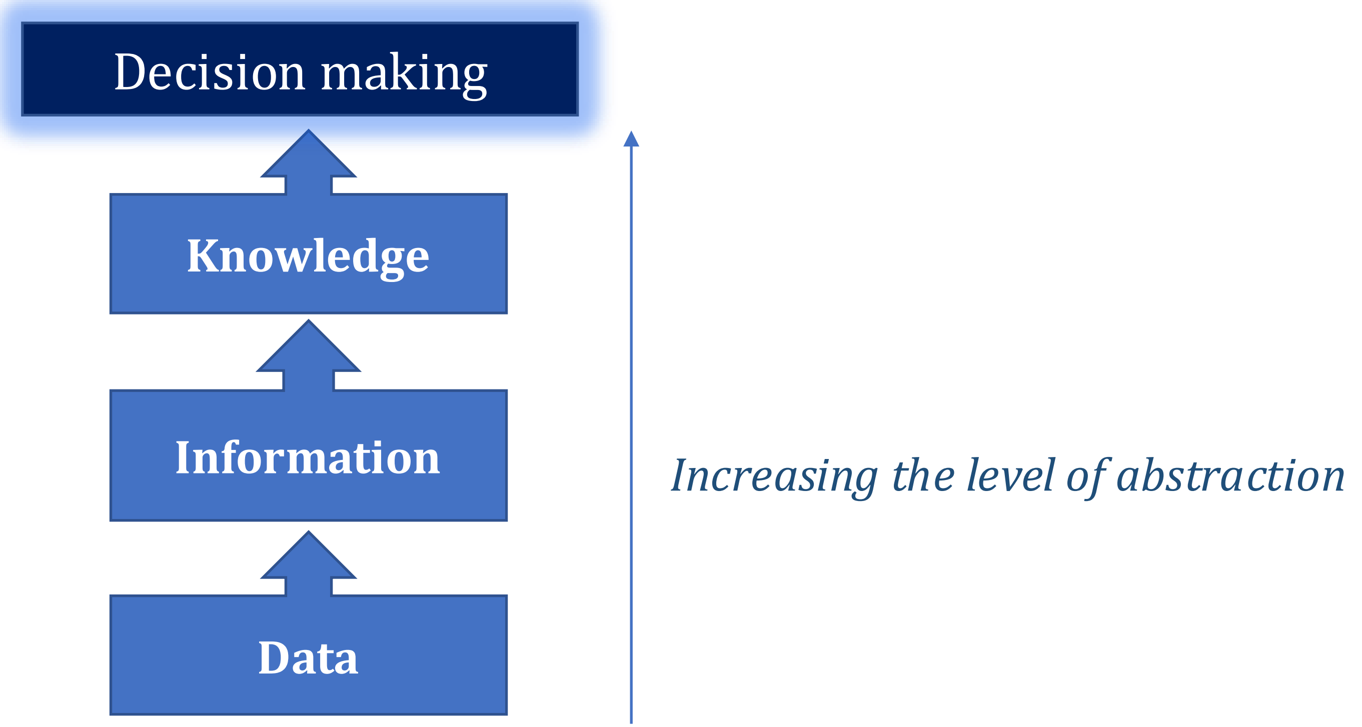
- What is *Analytics*?
  - Data.
  - Types of Analytics.
- What is *Text Analytics*?
  - How does Natural Language Processing work?
  - Supervised Machine Learning.
  - Use cases.
- Python “Open-Source” Libraries

How can we look at *data*?

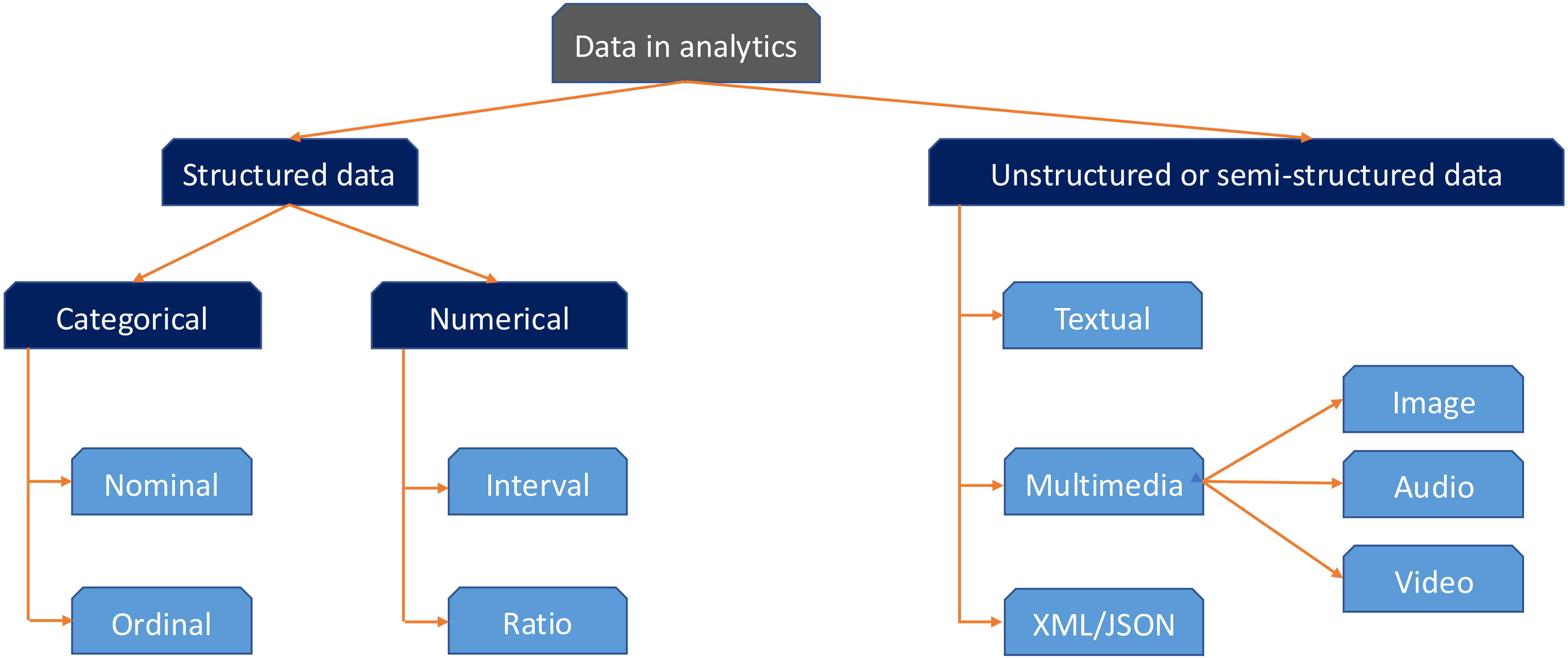


What is *data*?

- *data* (sg. **datum**)
  - a collection of facts usually obtained as the result of experiments, observations, transactions, or experiences
  - basically, we want to study a subject or an event, we decide what are the characteristics are of interest, and we measure these characteristics using a set of variables that will provide us with data in various forms: numbers, letters, words, images, voice recordings, etc.



A *data* taxonomy in analytics



What is analytics?

- techniques developed by mathematicians for practical problem solving, to apply math methods to real-world situations and everyday decision making
  - **Operations research** (O.R.): the application of advanced analytical methods to help make better decisions
  - **Analytics**: the scientific process of transforming data into insight for making better decisions
    - *the process of developing actionable decisions or recommendations for actions, based on insights generated from historical data, combining computer technology management science techniques, and statistics to solve real problems*

The Institute for Operations Research and Management Science (INFORMS), informs.org

What is analytics?

- *analytics* vs. *data science* ?
  - usually, by abuse of “notation”, *analytics* is used as a synonym for *data science*

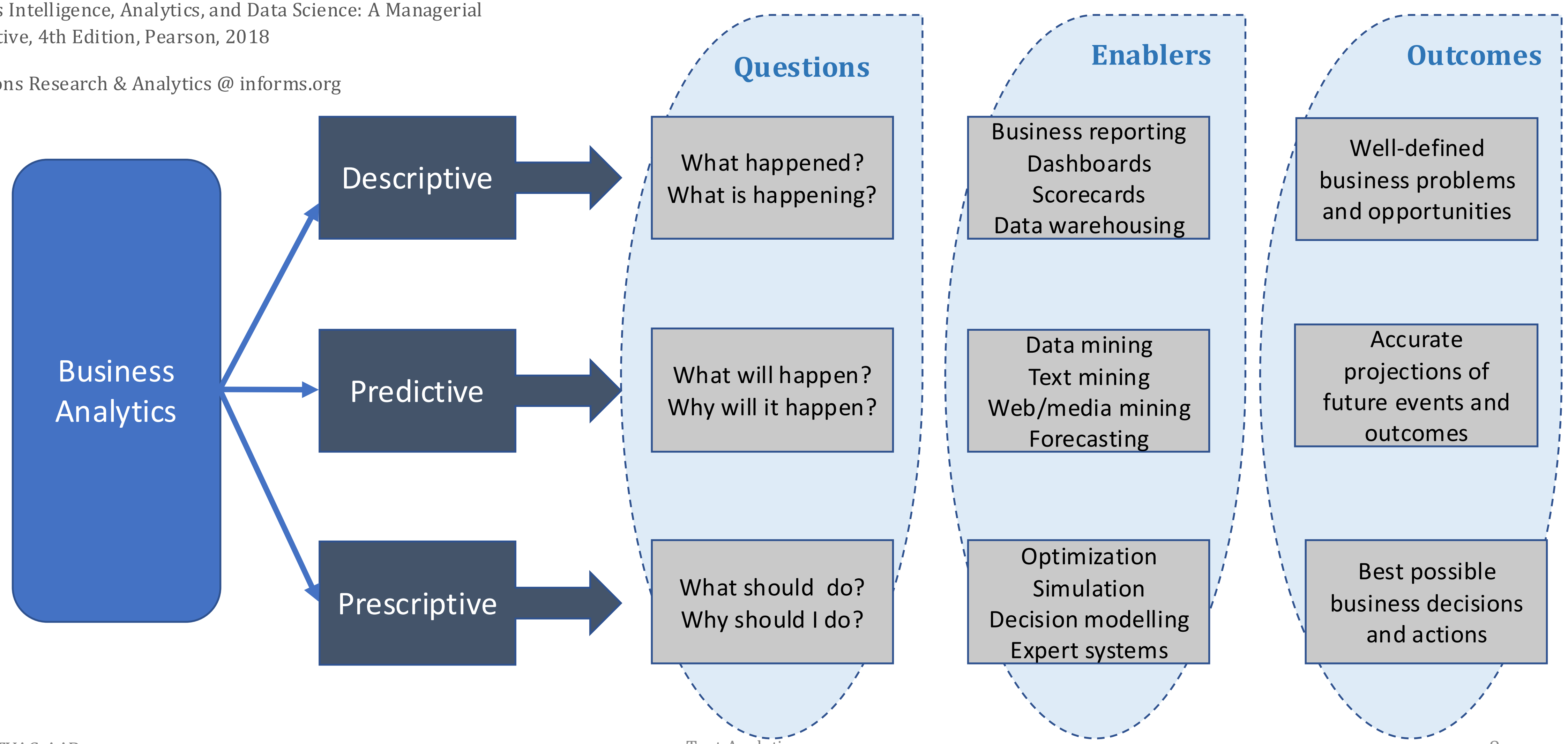
- why do we need *analytics*:
  - “…. to achieve various objectives, including in the field of business (*business objectives*), through the analysis of data collected in various ways, with the purpose of creating predictive models that will be used for forecasting and optimizing *business processes*, as well as for improving their performance.“
  - *analytics* (a.k.a ***advanced data analytics***) means applying specific scientific and mathematical methods to study & analyze various problems
  - this type of analysis is using complex algorithms, including of *machine learning* kind, besides the classical statistical techniques of data processing

INFORMS, Defining analytics: a conceptual framework, https://www.informs.org/ORMS-Today/Public-Articles/June-Volume-43-Number-3/Defining-analytics-a-conceptual-framework

Three categories of *analytics*

Adapted from: Ramesh Sharda, Dursun Delen, Efraim Turban, Business Intelligence, Analytics, and Data Science: A Managerial Perspective, 4th Edition, Pearson, 2018

Operations Research & Analytics @ informs.org



Three categories of *analytics*

- the *business analytics continuum*:

  1. Descriptive analytics: this is the type of analysis that provides insights into past events, based on previously collected data (*historical data*), e.g., studying the nature of the data, statistical modeling, data visualization, business intelligence."

  2. Predictive analytics: the type of analysis that provides insights into what might happen in the future; e.g., data mining, text/web mining, social media mining."

  3. Prescriptive analytics: that type of analysis that can be used to make decisions, with the purpose of providing actionable insights that can be used immediately, e.g., through optimization and simulation, Big Data approaches, trends, privacy, managerial considerations

INFORMS, Defining analytics: a conceptual framework, https://www.informs.org/ORMS-Today/Public-Articles/June-Volume-43-Number-3/Defining-analytics-a-conceptual-framework

What is text analytics?

Data

Text analytics use big sets of unstructured data (blogs, tweets, CRMs, claims or services notes, etc.)

Method

Train supervised ML algorithms that will associate words with specific meanings, to understand the semantic context of data.

Applicability

Sentiment analysis, text summarization/classification, topic modelling – Social Media, Sales & Marketing, Brand Monitoring, Customer Service, Security (email filtering).

… extract and decipher the meaning held within documents

Why text analytics

able to adapt business decision based on enormous amount of data.

understand massive amounts of complex data that are streaming into organizations, to *discover relationships between customer and business models*, to build knowledge and to make appropriate decisions

main pilar for *Customer Intelligence and Excellency*

… extract valuable information, to drive business excellency

How does Text Analytics / NLP work?

Information retrieval: gather relevant textual data from various sources (websites, social media platforms, customer surveys, reviews, etc.)

Data preprocessing:

Text cleaning: process of removing irrelevant characters, punctuation, special symbols and numbers from the dataset.

Convert text to lower case and stop-words (e.g. “the”, “is”, “with”, etc.) removal to ensure consistency.

Tokenization (breaks down the text into individual units (words) and stemming and lemmatization (normalize tokens to their root form).

Part of speech tagging (POS): assigning grammatical tags to tokens (noun, verb, adjective)

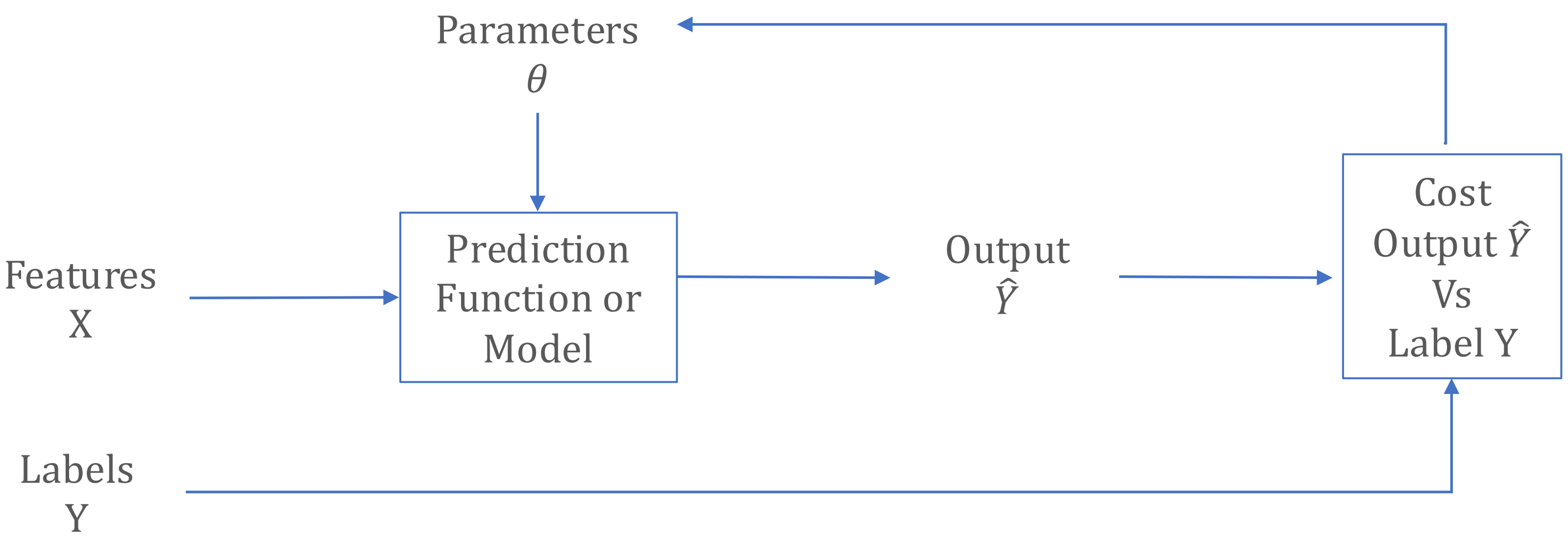
Text representation

Bag-of-words (BOW): represents text as a collection of unique words, each word becoming a feature and the frequency of occurrence represents its value. Term frequency-inverse document frequency (TF-IDF): calculates the importance of each word in a document based on its frequency or rarity across the entire dataset.

Data extraction:

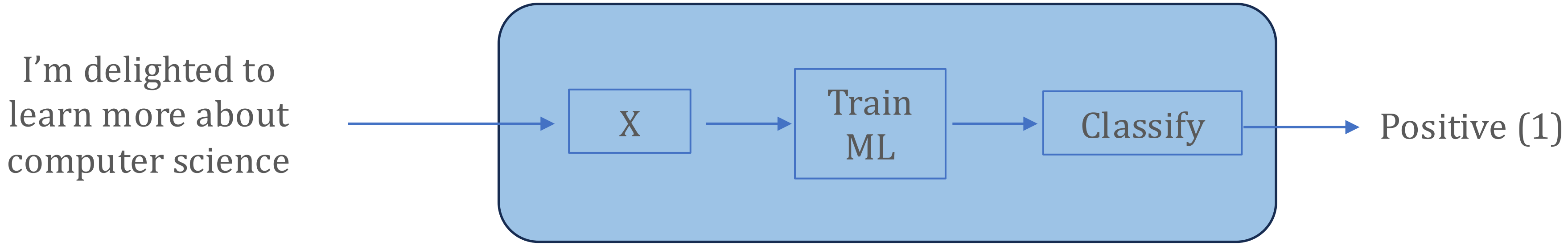
Apply one or more text mining techniques (sentiment analysis, text classification, name entity recognition). Popular algorithms: naïve bayes, logistic regression, convolutional neural networks, etc.

Supervised Machine Learning



- Input X goes into the NLP Model which will predict output $\hat{Y}$.
- During the model training we know for each set of features what are the correct labels.
- Based on a Objective Function (J) based on the Output vs Labels difference the model parameters will be updated (θ).
- More examples will de detailed.

Supervised Machine Learning



- Represent a quote, sentence as features by tokenization, stemming and lemmatization.
- With a trained model then we can classify the feedback (positive (1) or negative (0)).

Use cases

Sentiment Analysis:

- process of analyzing digital text to determine if the emotional tone of message is positive, negative or neutral. (AWS)
- why is important:
  - companies get consistent and objective results when analysing customers' opinions.
  - build better products and services
  - analyze at scale by mining information from a vast amount of unstructured data, such as emails, surveys, CRM records, product feedback.

Translation:

- the communication of the meaning of a source language in a target language text.
- example: Google translate.

Text summarization:

- Shortening a long text or paragraph to its concise summary

Use cases

Email filtering:

- filter out spam and malicious emails.
- examples: gmail, yahoo, microsoft etc.

Smart assistants:

- recognize pattern in speech using voice recognition.
- examples: Apple Siri, Amazon Alexa.

Conversational AI (Chatbots):

- technology that enables automatic conversation between computers and humans.
- rely on NLP and intent recognition to understand user queries, mine in their training data and generate relevant response.
- examples: ???

Python Open Source libraries for Text Analytics

- NLTK (Natural Language Toolkit)
  - *Leading platform for building python scripts to work with human language data.*
  - Suite of text processing libraries for classification, tokenization, stemming, tagging, parsing, etc

```
>>> import nltk
>>> sentence = """At eight o'clock on Thursday morning
... Arthur didn't feel very good."""
>>> tokens = nltk.word_tokenize(sentence)
>>> tokens
['At', 'eight', "o'clock", 'on', 'Thursday', 'morning',
'Arthur', 'did', "n't", 'feel', 'very', 'good', '.']
>>> tagged = nltk.pos_tag(tokens)
>>> tagged[0:6]
[('At', 'IN'), ('eight', 'CD'), ("o'clock", 'JJ'), ('on', 'IN'),
('Thursday', 'NNP'), ('morning', 'NN')]
```

  - Use cases: text classification, information extraction/ summarization, language understanding, text generation, chatbot etc.

Python Open Source libraries for Text Analytics

- Gensim
  - *Well suited for handling large corpora and for training models for topic modeling, document similarity analysis.*

  - Pro:
    - Efficiency and scalability: can handle large text datasets efficiently.
    - Topic modelling: excels in topic modeling tasks allowing user to discover hidden topics within a collection of documents.
    - Word embeddings: Word2Vec and Doc2Vec algorithms.
    - Easy-to-Use interface.
    - Compatibility: fully compatible with other Python libraries like NumPy, SciPy, Sci-kit-learn.

  - Cons:
    - Sparse Documentation.
    - Limited Deep Learning support comparing with TensorFlow or PyTorch.

  - Use cases: topic modelling – uncovering topics in large document collections, document similarity, text summarization, information retrieval.

- CoreNLP
  - *NLP library developed by Stanford Natural Language Processing Group.*
  - *Supports a wide range of NLP tasks, including part-of-speech tagging, named entity recognition, sentiment analysis, etc.*

  - Pro:
    - Cover a wide range of NLP tasks making it versatile choice for various natural language processing.
    - Integrated pipeline allowing users to perform multiple NLP tasks without the need for separate tools.
    - Multilingual support.
    - Pre-trained models.
    - Active Development the library being actively maintained and updated by Stanford NLP Group.

  - Cons:
    - Resource intensive.
    - Dependency on Java and limited customization.

  - Use cases: information extraction, sentiment analysis, document summarization, question answering, etc.

Python Open Source libraries for Text Analytics

- Auxiliary libraries:
  - ***Numpy*** *– offers comprehensive mathematical functions, random number generators, linear algebra, etc.*
    - *https://numpy.org*
  - ***Scipy*** *- provides algorithm for optimization, integration, interpolation, eigenvalue problems, differential equations, statistics.*
    - *https://scipy.org*
  - ***Scikit-learn*** *– module for machine learning built on top of SciPy. Offer various functions for classification, regression, model validation, etc*
    - *https://pypi.org/project/scikit-learn/*
  - ***Tensorflow*** *– library for machine learning and artificial intelligence, used mainly for training and inference of neural networks.*
    - *https://tensorflow.org.*
  - ***Matplotlib*** *– plotting library offers object-oriented API for embedding plots.*
    - *https://matplotlib.org*
  - ***Pandas*** *- library used for data manipulation and analysis. Offers data structures and operation for manipulating numerical tables and time series.*
    - *https://pandas.pydata.org*